

Time: 1.0 hr

Sub: IT

F.M 26

1. Why Computer is known as versatile machine? (3)
2. Convert $(132)_8$ into Binary. (3)
3. What is Boolean algebra? Prove De.Morgan's Theorem. (8)
4. Define operating system. Explain its function. (12)

Subject: Mathematics

1. Find the nth derivative of $\sin(ax+b)$ 3
2. Find coordinates of vertex, focus and length of latus rectum of $x^2=12y$3
3. Find the reduction formula of $\int \sin^n x dx$ 8
4. State and prove Leibnitz's Theorem. If $y=e^{at}u^{-1}x$ prove that $(1+x^2)y_2+(2x-a)y_1=0$12

Or Integrate $\int_0^{\frac{\pi}{2}} \frac{dx}{1+\cos x}$

Time: 1.0 hr Sub: C programming

Attempt all Questions

1. Write characteristics of algorithm3
2. Define keywords supported by C3
3. Differentiate between while and do while loop with example3
4. Describe structure of C program with block diagram8
5. Write C program to display prime numbers between 1 to 1008
6. Write C program to check whether entered number is palindrome or not?8
7. Write C program to display list of numbers in descending order12

FM: 20

PM: 9

Far Western University

Department of CSIT

Time: 1.0 hr

Midterm Examination 2079

Attempt all Questions (5*4)

1. What can you do to create the good first impression on your classmate?
2. How has technically affected your life? technology
3. What makes food taste good? Describe any one dish you like.
4. Do you think sponsoring games and events promote business? If yes how?

FM: 20

PM: 9

Time: 1.0 hr

Sub: Physics

Midterm Examination 2080

1. Explain the significance of Q-point on a dc load line. (3)
2. What is Zener diode? Explain how a Zener diode can be used as voltage regulator. (7)
3. (a) What is Barkhausen criterion for oscillation? Explain the working of Hartely oscillator. (7)
- (b) Differentiate between p-n junction diode and zener diode. (7)

FAR WESTERN UNIVERSITY
Semester End Examination-2080
English Grammar and Composition (CSIT.111)

Faculty: Science and Technology (CSIT)

Level: Undergraduate

Semester: First

Full Marks: 100

Time: 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.

Group – 'A'

Attempt all questions (Very short answer questions)

8×3=24

1. What is the impact of casual clothing to a meeting?
2. Differentiate between supertasters and nontasters?
3. Explain about the economy of Second life?
4. What image did Kelly want to change by climbing Half Dome Mountain in Yosemite?
5. What is unique about Masdar city?
6. What are the advantages of social networking sites?
7. How are parents responsible for their child to be a successful athlete?
8. How can we use technology in our daily lives?

Group – 'B'

Attempt any five questions. (Short answer questions)

5×8=40

9. Explain the merits and demerits of social networking sites?
10. How Beatrice got success by dint of a small help?
11. Will Masdar be a success? Explain.
12. What are pros and cons of advertising?
13. Why do you think people take risk? Explain.
14. Have you ever received something that made you very happy? How?

Group – 'C'

Attempt any three questions. (Long answer questions)

3×12=36

15. What is bystander's effect? Explain in detail.
16. What positive and negative effects can technology have on people's lives?
17. How can you create a good first impression on your classmate? Discuss.
18. Describe your favourite dish and explain if it has cultural importance to you.

*face experience
frequently smile*

FAR WESTERN UNIVERSITY
Semester End Examination-2080
Information Technology Fundamentals (CSIT.112)

Faculty: Science and Technology (CSIT)

Level: Undergraduate

Full Marks: 100

Semester: First

Time: 3hrs.

*Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.*

Group – 'A'

Attempt all questions (Very short answer questions)

8×3 = 24

1. ✓ "Computer is diligent machine". Justify your answer.
2. Define database and DBMS.
3. ✓ Differentiate between soft copy and hard copy output.
4. ✓ Define half adder. ✓
5. Why is NAND gate called universal gate?
6. ✓ What is computer crime?
7. ✓ Convert (BCA) into octal system. *Handwritten note: Binary to octal.*
8. What is search engine? List any two examples.

Group – 'B'

Attempt any five questions. (Short answer questions)

5×8=40

9. ✓ Differentiate between third and fourth generation computers.
10. ✓ Define number system. Subtract $(1010)_2$ from $(111)_2$ using 1's and 2's complement method.
11. Explain the memory hierarchy in brief.
12. ✓ What do you mean by mobile computing? Describe its advantages.
13. ✓ Define simplex, half duplex and full duplex channel of communications with examples.
14. ✓ Differentiate between GUI and CUI with example.

P.T.O.

Group – 'C'

Attempt any three questions. (Long answer questions)

3×12 = 36

15. Why do we need computer network? Discuss different types of network topologies along with their merits and demerits.
16. Draw a block diagram of computer system and explain the main components of basic computer system.
17. What is a logic gate? Explain three basic logic gates with diagram and truth tables.
18. What is programming language? Describe the various types of programming languages with their merits and demerits.

$$F = (A + B) \text{ OR } \rightarrow \text{OR}$$

$$F = (A \cdot B) \text{ AND } \rightarrow \text{AND}$$

$$F = \frac{1}{2}$$

32 16 8 4 2 1

FAR WESTERN UNIVERSITY
Semester End Examination-2080
Calculus and Analytical Geometry (CSIT.113)

(62)

Faculty: Science and Technology (CSIT)

Level: Undergraduate

Semester: First

Full Marks: 100

Time: 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.

Group -- 'A'

Attempt all questions (Very short answer questions)

8×3 = 24

1. Show that the sequence $\langle \frac{n+1}{n} \rangle$ is monotonically decreasing.

2. Show that the series $\sum_{n=1}^{\infty} \frac{1}{9} \left(\frac{1}{3}\right)^{n-1}$ is convergent. (3)

3. If $u = x^2 + y^2 + z^2$ show that $xu_x + yu_y + zu_z = 2u$ (3)

4. Find the n^{th} derivative of the function $\sin(ax+b)$ (3)

5. Using properties of definite integral, show that (3)

$$\int_0^{\pi} xf(\sin x)dx = \frac{\pi}{2} \int_0^{\pi} f(\sin x)dx$$

6. Find the reduction formula of $\int x^n e^{ax} dx$ (2)

7. Find the equation of parabola whose vertex is $(-1,1)$ and directrix is $y=3$

8. Find the area of the parallelogram whose adjacent sides are determine by the vectors $\vec{u} = \vec{i} + \vec{j} + \vec{k}$ and $\vec{v} = \vec{i} + 2\vec{j} - \vec{k}$

Group -- 'B'

Attempt any five questions. (Short answer questions)

5×8=40

9. Test the convergency of the p-series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ by integral test (5)

10. Find co-ordinates of center, vertex, eccentricity, length of major and minor axis, foci and length of latus rectum of the ellipse

$$3x^2 + 4y^2 - 12x - 8y + 4 = 0$$

11. Find the reduction formula of $\int \cos^m x \sin x dx$ and hence evaluate $\int \cos^2 x \sin 3x dx$

Handwritten notes and calculations:
 $\int \cos^m x \sin x dx = -\frac{\cos^{m-1} x}{m-1} + \int \cos^{m-2} x \sin x dx$
 $\int \cos^2 x \sin 3x dx = \int \cos^2 x \sin x \cdot 3 \cos^2 x dx = 3 \int \cos^4 x \sin x dx$
 $\int \cos^4 x \sin x dx = -\frac{\cos^3 x}{3} + \int \cos^2 x \sin x dx$
 $\int \cos^2 x \sin x dx = -\frac{\cos^3 x}{3} + \int \cos x \sin x dx = -\frac{\cos^3 x}{3} - \frac{\cos^2 x}{2} + C$

12. Evaluate the following integrals

i. $\int_1^3 \int_0^2 (4-y)^2 dy dx$

ii. $\int_{-1}^1 \int_0^{\sqrt{1-x^2}} dy dx$

iii. $\int_0^1 \int_0^1 \int_0^1 xyz dx dy dz$

iv. $\int_0^{2\pi} \int_0^1 \int_{-1/2}^{1/2} (r^2 \sin^2 \theta + z^2) dz r dr d\theta$

13. i. If $V = (x^2 + y^2 + z^2)^{-1/2}$ prove that $V_{xx} + V_{yy} + V_{zz} = 0$

ii. If $U = \log(x^3 + y^3 + z^3 - 3xyz)$ show that $\frac{\partial^2 x}{\partial x^2} + \frac{\partial^2 y}{\partial y^2} + \frac{\partial^2 z}{\partial z^2} = \frac{-3}{(x+y+z)^2}$

14. If $\vec{v} = 2\vec{i} + 10\vec{j} - 11\vec{k}$ and $\vec{u} = 2\vec{i} + 2\vec{j} + \vec{k}$ then find the followings

i. $\vec{u} \cdot \vec{v}$ and $\vec{u} \times \vec{v}$

ii. the cosine of the angle between the vectors $\vec{u}$ and $\vec{v}$

iii. The vector projection of $\vec{u}$ on $\vec{v}$

Group - 'C'

Attempt any three questions. (Long answer questions)

3x12 = 36

15. State ratio test and use it to test the convergence of the following series.

i. $\sum \frac{n^2(n+1)^2}{n!}$

ii. $\sum_{n=1}^{\infty} \frac{n^2}{3^n}$

16. Define beta and gamma function. Use gamma function prove that

i. $\int_0^{\pi/6} \cos^4 3\theta \sin^2 6\theta d\theta = \frac{5\pi}{192}$

ii. $\int_0^{2a} x^5 \sqrt{2ax - x^2} dx$

17. State and prove Leibnitz's theorem. Use it, If $y = \sin^{-1} x$ prove the followings

i. $(1-x^2)y_2 - xy_1 = 0$

ii. $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} + n^2 y_n = 0$

18. Find the parametric equation of the line through the given point and parallel to the given vector. Also find the perpendicular distance from a point to a line in a space. Find a vector parallel to the intersection of the planes $x-2y+4z=2$ and $x+y-2z=4$ and also find the parametric equation of this line.

FAR WESTERN UNIVERSITY
Semester End Examination-2080
Electronic Principles (CSIT.114)

Faculty: Science and Technology (CSIT)

Level: Undergraduate

Semester: First

Full Marks: 100

Time: 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.

(49)

Group – 'A'

Attempt all questions (Very short answer questions)

8×3 = 24

1. Define Knee voltage? Show it in graph. (5)
2. Why is biasing of transistor required? (1)
3. Why DE-MOSFET is called ON MOSFET? (2)
4. Differentiate between FET and BJT? (2)
5. Define dc load line? Give its importance.
6. State the condition under which feedback amplifier works as oscillator? (1)
7. Differentiate between ideal voltage source and current source. (2)
8. Define pinch off voltage? (2)

Group -- 'B'

Attempt any five questions. (Short answer questions)

5×8=40

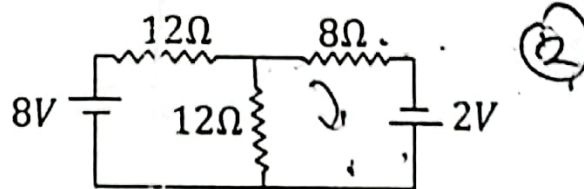
9. What is Zener diode? Show how a Zener diode can be used to regulate voltage. (5)
10. State and explain Thevenin's theorem? (5)
11. Define operational amplifier? Show that how an operational amplifier can be used as adder and differentiator? (5)
12. Draw the circuit diagram of a Hartley oscillator and explain its action?
Write down the uses of oscillator.
13. Define regulated power supply. Draw the circuit diagram of series voltage regulator and explain its operation. (5)
14. Define the terms
(a) Extrinsic Semiconductor
(b) Photodiode
(c) Virtual ground. (6)

(3+3+2)

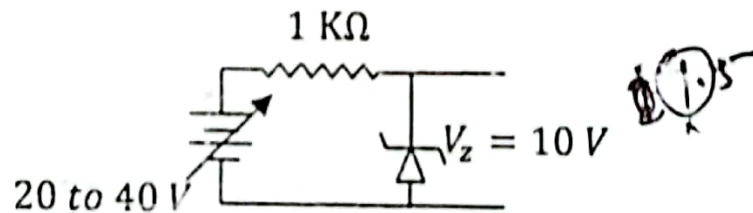
Attempt any three questions. (Long answer questions)

3×12 = 36

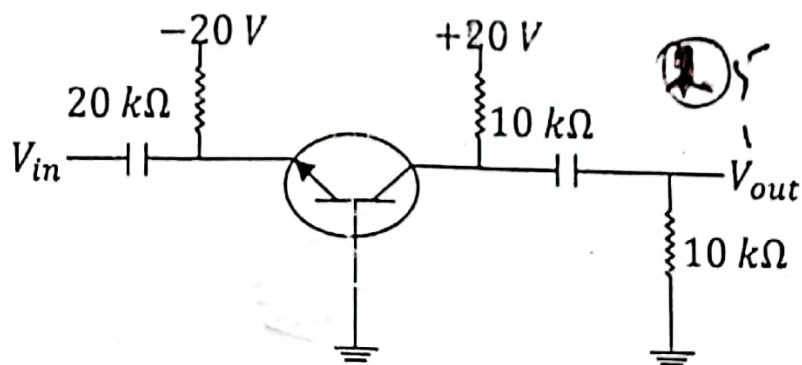
15. (a) Draw the static and transfer characteristic of JFET and explain the different portion of the characteristics. (8)
 (b) Using Kirchoff's laws, determine value of current in each resistors. (4)



16. (a) What is rectifier? Explain the use of PN junction diode as full wave rectifier? (8)
 (b) Use ideal approximation to calculate the output voltage in the figure below. What is the minimum Zener current? (4)



17. (a) Discuss common emitter characteristics with proper theory and circuits? (8)
 (b) Calculate input resistance, output resistance and voltage gain for the given circuit. Assume $\alpha = 1$ and $V_{BE} = 0$. (4)



18. (a) Describe direct coupled two stage amplifier and obtain the relation for total voltage gain. (8)
 (b) Calculate the α - parameter of a transistor if $\beta = 99$. (4)

FAR WESTERN UNIVERSITY
Semester End Examination-2080

Programming Fundamentals and C Programming (CSIT.115)

Faculty: Science and Technology (CSIT)

Level: Undergraduate

Semester: First

Full Marks: 100

Time: 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.

Group – 'A'

Attempt all questions (Very short answer questions)

1. Mention the key features of the C programming language? — (2)
2. Differentiate between = and == symbols in C programming with examples? (2)
3. Describe the general form of function in C? (3)
4. What are static variables? Explain with examples. (2)
5. What is the main difference between the Compiler and the Interpreter? (2)
6. Differentiate between getch() and getche() with syntax and examples. (2)
7. What are the limitations of scanf() and how can it be avoided? (1)
8. Explain ternary operator in c with examples. (1)

Group – 'B'

Attempt any five questions. (Short answer questions)

9. Why programming tools and techniques are necessary for developing program. Explain flowchart with symbol. (4)
10. What is data type? Explain various types of data types with examples. (4)
11. Write rules followed in declaring variable. (6)
12. Write an algorithm to find sum of n natural numbers and also draw flowchart. (10)
13. Write c program to perform arithmetic operation on two operand depending on type of operator entered by using switch statement. (10)
14. Write c program to check whether entered number is Armstrong number or not. (10)
15. What is recursive function? Write c program to find factorial of entered number using recursion. (10)

P.T.O.

⑤

Group - 'C'

Attempt any three questions. (Long answer questions)

3×12=36

15. Define string with example. Explain strcmp() function with its syntax.
Write c program to enter name and address of n employees and sort them in alphabetical order according to name.
16. Explain various types of user defined functions with examples.
17. Write c program to read n students record with the fields Roll No., Name, Faculty and DOB (dd, mm, yy) and display the record of students belonging to "CSIT" faculty.
18. What is array? Describe types of array with declaration syntax and examples. Write c program to enter n numbers and search given number in the list.

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